ARTIFICIAL INTELLIGENCE – LENOVO LEAP

An Internship Report For

Course code and Course Title: CSA-321 and Internship

Credits: 04

Submitted in partial fulfillment of Bachelor's Degree

Bachelors of Computer Applications

by:

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SPES's Goa Multi-Faculty College Dharbandora-Goa

Computer Application

GOA UNIVERSITY

Date: October 2025

Examined by: **Seal of the School/ College Dept**

DECLARATION BY STUDENT

I hereby declare that the report titled "AI Internship – Lenovo Leap Program" is based on the internship carried out by me in the Department of Computer Applications, Goa Multi-Faculty College, Dharbandora, under the mentorship of Mr. Dnyandev Khadapkar. This report is a record of my original work and has not been submitted elsewhere for the award of any degree or diploma by me. Further, I understand that Goa University or its authorities/College will not be responsible for the correctness of the observations, experimental, or other findings presented in this internship report. I also authorize the College authorities to upload this internship report on the concerned repository or any other platform as per UGC regulations, and to make it publicly accessible as needed.

Kaushik Ravi Advekar
Signature & Name of Student
Seat No.: 40206

Date:

Place: Goa Multi-Faculty College, Dharbandora-Goa

COMPLETION CERTIFICATE

This is to certify that the internship report titled "AI Internship – Lenovo Leap Program" is a bonafide work carried out by Mr. Kaushik Ravi Advekar under my mentorship in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications (BCA) in the discipline of Computer Applications at the Department of Computer Applications, SPES'S Goa Multi-Faculty College, Dharbandora

MR. DNYANDEV KHADAPKAR

Signature and Name of Internship Faculty Mentor

Date:

Signature of HOD of College

Department of IT & Computer Applications

Stamp

Date:

Place: Goa Multi-Faculty College, Dharbandora-Goa.

College: SPES'S Goa Multi-Faculty College, Dharbandora

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OFFER LETTER

Internship Offer Letter

Date: 21 July 2025.

Dear Kaushik Advekar

We are pleased to inform you that you have been selected for the 8-week **Lenovo Leap: NextGen Scholar Internship Program**—a CSR initiative by **Lenovo**, conducted in collaboration with the **BharatCares** Foundation and supported by the **Directorate of Higher Education, Goa.**

The internship is scheduled from **July to August 2025** and offers a unique opportunity to work on real-time projects in the field of emerging technologies, with a focus on **Artificial Intelligence(AI), Data Analytics(DA) and Web development(WD).**

As an intern, you will engage in a carefully curated learning journey that includes:

- Project-Based Learning for real-world exposure
- Structured Learning Modules on AI ,WD and DA
- Expert-led Masterclasses and industry-led interactive webinars
- Mentorship and Doubt-Clearing Sessions

This internship is designed to help you build key technical and professional competencies, including:

- Content Knowledge of selected internship
- Problem Solving and Critical Thinking
- Innovation, Collaboration, and Effective Communication

On successful completion of the internship and active participation throughout the program, you will be awarded an Internship Completion Certificate.

We are excited to have you on board and look forward to your enthusiastic participation in this transformative experience.

Chandni Patel

Chandni Nagar Patel

Associate Vice President - People & Culture

COMPLETION OF THE INTERNSHIP CERTIFICATE

Certificate of Completion

This certificate is proudly presented to

Kaushik Advekar

for completing 60+ hours of internship in **Artificial Intelligence** in the Lenovo Leap : NextGen Scholar Program conducted by BharatCares in collaboration with Lenovo.

Wishing you all the best in your future endeavors!

Shailendra Katyal
(VP and MD,
Lenovo India Region)

T M Narasimhan
(Managing Director,
India MBG, Lenovo India Region)

Bhomik Shah
(Founder and
CEO, BharatCares)

Lenovo LEAP | motorola A Lenovo company | Bharat Cares

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to Lenovo Leap for providing me with the opportunity to complete my internship in the field of Artificial Intelligence. This internship has been a valuable learning experience, allowing me to gain practical knowledge and hands-on exposure to real-world AI concepts.

I would also like to thank my mentors and the entire Lenovo Leap team for their constant guidance, support, and encouragement throughout the internship. Their insights and feedback helped me enhance my technical skills and better understand the applications of AI in modern technology.

I am deeply thankful to my college faculty mentor **Dynandev Khadapkar** and the Department of **IT & Computer Applications** for their continuous guidance, encouragement, and for providing this opportunity to apply classroom knowledge in a real-world context. The learning and exposure gained through this internship have significantly contributed to my academic and professional growth.

I also express my sincere appreciation to the **Principal, faculty members, and SPES's Goa Multi-Faculty College** for their encouragement and for providing this platform to gain hands-on industry experience.

Finally, This experience has been a remarkable milestone in my learning journey, and I will always cherish the lessons and memories it brought along.

EXECUTIVE SUMMARY

This report presents an overview of the **Lenovo LEAP Artificial Intelligence Internship**, a structured 8-week virtual program consisting of 60 hours of guided learning, assignments, and project-based activities. The internship was designed to introduce participants to the fundamentals of **Artificial Intelligence (AI)** and **Machine Learning (ML)** through weekly learning modules, hands-on tasks, and practical deliverables.

The program provided a comprehensive understanding of core AI concepts, including machine learning workflows, basic algorithms, chatbot development, and the creation of concept notes for AI-based solutions. Weekly instructional videos and assignments allowed me to gradually build technical skills while applying them through tasks such as **chatbot creation**, preparing **PowerPoint presentations**, submitting **deliverables**, and developing a **final AI project idea**.

The internship also presented certain challenges, such as managing weekly deadlines, understanding complex ML topics with limited prior exposure, and adapting to the self-paced online learning format. However, these experiences significantly contributed to my growth, adaptability, and discipline.

Overall, the Lenovo LEAP AI Internship was an enriching learning experience that enhanced my foundational knowledge of AI and ML while improving my technical, analytical, and communication skills. The insights, tasks, and final project ideation have provided a strong base for pursuing future academic and professional opportunities in the field of Artificial Intelligence.

CHAPTER 1: ORGANIZATION/COMPANY

1.1 Birds-eye-view

Lenovo Leap is an educational initiative under Lenovo India that focuses on empowering students and young professionals with advanced technological skills. The program is designed to provide hands-on exposure to emerging technologies, especially Artificial Intelligence (AI), Data Analytics, and Web Development, preparing participants for real-world industry challenges.

The initiative serves as a bridge between classroom learning and practical application, allowing interns to gain experience with live projects, modern AI tools, and mentorship from industry professionals. Lenovo Leap reflects Lenovo's global vision of innovation, education, and social responsibility, while also promoting inclusivity, gender equality, and sustainable practices within technology education.

During the internship, I observed how the organization creates a structured learning ecosystem where interns are actively engaged in AI-based projects, encouraged to explore innovative solutions, and prepared to meet industry standards.

1.2 Products / Services

Although Lenovo Leap is primarily an educational and skill-development program,
it provides services that mirror real-world technology applications:

1. AI Training Modules: Interns are trained on Artificial Intelligence fundamentals, including Machine Learning, Natural Language Processing, and AI-powered automation tools.

2. Hands-On Project Experience: Participants work on practical projects, developing AI solutions to real-world problems.

3. AI Lab Access: The program offers access to AI laboratories and cloud-based tools for experimentation and learning.

4. Mentorship & Guidance: Industry experts mentor interns, offering technical insights and feedback to help refine their skills.

5. Certification & Recognition: Interns receive certifications for completing the program and have opportunities to showcase their work in state-level or organizational events.

These services ensure that interns not only understand theoretical concepts but also gain the practical knowledge and experience required to apply AI solutions effectively.

Sections within the Organization The Lenovo Leap internship program is organized into multiple functional sections to facilitate structured learning and skill development:

1. Training & Development Section:
   - Conducts workshops, tutorials, and guided sessions in AI, Data Analytics, and other technologies.
   - Prepares interns for project-based learning.

2. Research & Innovation Section:
   - Focuses on AI experimentation and development of new tools or techniques.
   - Encourages interns to explore innovative AI applications in real-world scenarios.

3. Mentorship & Evaluation Section:
   - Provides one-on-one mentorship and monitors progress throughout the internship.
   - Offers constructive feedback to enhance skills and understanding.

4. Project Showcase & Recognition Section:
   - Organizes events where interns present their projects and solutions.
   - Highlights top performers and acknowledges innovative contributions at the state or program level.

During my internship, I primarily worked under the Training & Development Section, where I learned to implement AI concepts, use practical tools, and understand their applications in real projects

1.3 Sections Within the Organization

The Lenovo LEAP program operates under **Lenovo India**, which consists of several functional divisions that work together to deliver technology solutions, innovation, and community-driven skill development initiatives. The key sections relevant to the LEAP internship are as follows:

1. Corporate Social Responsibility (CSR) Division

This section oversees Lenovo's community-oriented programs, including the Lenovo LEAP skilling initiative. It focuses on designing and implementing projects that promote digital literacy, youth skill development, and technological empowerment.

2. Lenovo LEAP Program Management Team

This section is responsible for planning, coordinating, and managing the internship structure. It handles weekly content release, communication with interns, monitoring progress, and ensuring timely completion of assignments and deliverables.

3. Learning & Development (L&D) Unit

This team designs the instructional modules for AI, Machine Learning, Data Analytics, and Web Development. It develops training material, video lectures, tutorials, and exercises for participants.

4. Technical Mentorship & Support Team

This section provides technical guidance, clarifies doubts, assists with project-related queries, and supports participants in understanding AI/ML concepts and tools used in the internship.

5. Evaluation & Assessment Unit

This unit assesses weekly submissions, project deliverables, presentations, and the final project idea. It ensures that the learning outcomes are effectively met and provides feedback to improve performance.

6. Partner Organization – BharatCares

BharatCares acts as the implementation partner for Lenovo LEAP. It coordinates logistics, monitors student engagement, sends updates, and ensures smooth execution of the internship at the ground level

CHAPTER 2:TASK(S) HANDLED

During my 8-week AI internship at Lenovo Leap, I undertook a series of structured learning modules, projects, and hands-on assignments that enhanced my understanding of Artificial Intelligence, Machine Learning, Deep Learning, NLP, and Generative AI.

2.1 Introduction to AI: Week 1

Introduction to Artificial Intelligence in week 1

- Studied core AI concepts including AI vs. Machine Learning, system components, and logic-based decision-making. Learned to design AI agent goals, create flowcharts, and build simple models using Python or no-code tools.
- Explored types of AI, decision trees, and AI learning processes through masterclasses, case studies, and interactive assignments.
- Making an AI Agent: Learned about different types of AI agents, their working, and real-world applications. Explored how AI agents interact with their environment to make decisions and solve problems.

2.2 Machine Learning: Week 2

- Week 2: Machine Learning Fundamentals: Gained foundational knowledge of supervised, unsupervised, and reinforcement learning. Learned stages of ML process including data preparation, model training, and evaluation.

- Python Fundamentals: Covered variables, data types, loops, conditionals, functions, and data handling in Python.

- **Linear Regression Model Project: 11**
    - Cleaned and prepared datasets for modelling.
    - Implemented and trained a Linear Regression model using Python in Google Colab.
    - Evaluated model performance with suitable metrics.

Visualize actual pricing of houses and area of house prices

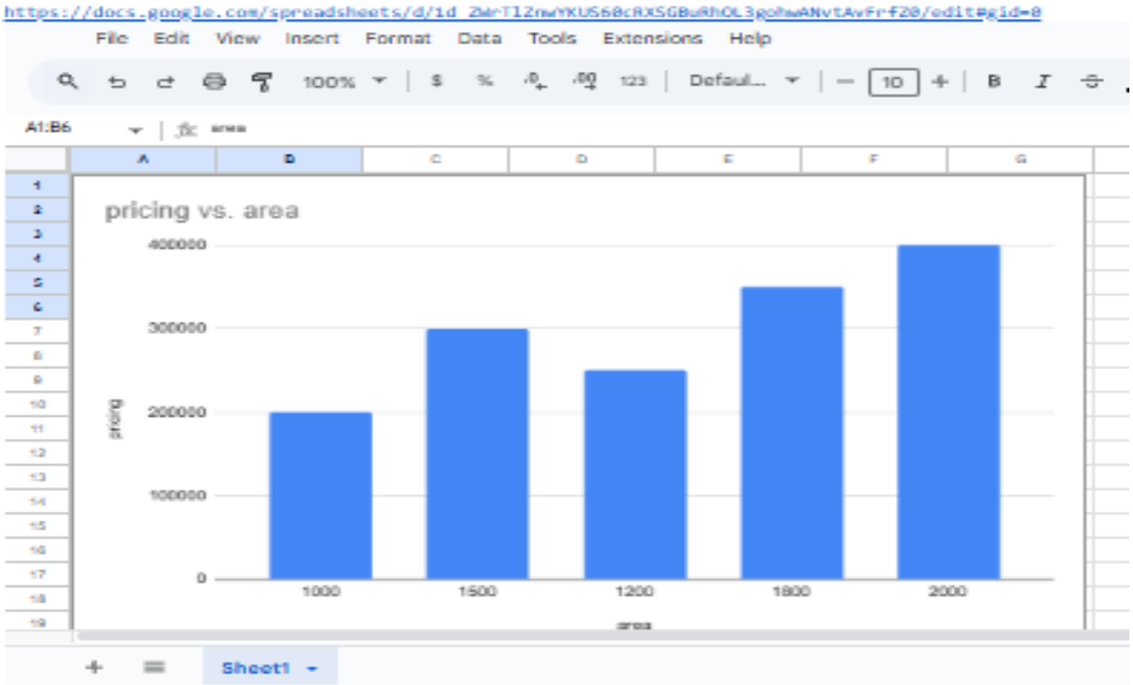
```
[4]
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
```

```
[9]
data = pd.read_csv('house_pricing.csv')
X = data[['area']]
Y = data['pricing']
data.head()
```

	area	pricing
0	1000	200000
1	1500	300000
2	1200	250000
3	1800	350000
4	2000	400000

```
from google.colab import sheets
sheet = sheets.InteractiveSheet(df=data)
```

https://docs.google.com/spreadsheets/d/1d_ZWrTlZmwYKUS68cRXSGBuRhOL3goHwANvtAvFrf20/edit#gid=0



2.3 Introduction to Deep Learning: Week 3

- Studied artificial neural networks (ANNs), multilayer perceptrons (MLPs), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs).
- Learned how deep learning models process information and their applications in AI systems.

2.4 Natural Language Processing (NLP): Week 4

- Learned key NLP concepts and techniques to analyze text data, recognize language patterns, and extract insights.
- Understood text representation, context learning in machines, and applications like chatbots, language translation, and content summarization.

2.5 Project Assignment – Designing a Specialized AI Assistant Persona: Week 5

- Goal: Design a specialized AI assistant with a defined purpose, persona, and knowledge domain using prompt engineering and Generative AI tools.

- **Tasks:**

1. Conceptual Design: Defined assistant's niche, target audience, purpose, personality, tone, and key knowledge areas.
2. Prompt Engineering: Created 5–7 detailed meta-prompts guiding assistant behavior.
3. Sample Interactions: Wrote 5 user prompts and ideal responses.
4. Testing & Refinement: Simulated assistant on platforms like ChatGPT, iteratively refined prompts, and analyzed results.

- **Deliverable**s: Project report containing AI assistant profile, prompt engineering, sample interactions, testing analysis, reflections, and ethical considerations

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2.6 Build a Chatbot using Botpress: Week 6

- Developed a functional chatbot using Botpress, learning about:
- Chatbot types: rule-based and AI-based
- Conversational flow design
- Nodes, intents, entities, and smooth conversation transitions
- Applied hands-on skills to deploy a basic real-world chatbot.

2.7 Concept Note Submission: Week 7

- Submitted a concept note for the Capstone Project, aligning AI solutions with Sustainable Development Goals (SDGs).
- Explored potential AI applications for:

SDG 3 (Health & Well-being): AI for mental health support and early disease detection

SDG 4 (Education): AI-powered personalized learning and quiz Generators

SDG 5 (Gender Equality): AI tools for bias detection and awareness

SDG 8 (Economic Growth): Resume builder and career guidance AI Agents

SDG 11 (Sustainable Cities): Smart traffic prediction, waste management AI

SDG 13 (Climate Action): Carbon footprint tracking and eco-friendly Habit promotion

- Submitted structured concept note with title, problem statement, SDG relevance, AI solution, tools, target audience, and expected impact.

Link Of the Concept notes

https://docs.google.com/presentation/d/13j-EgjFmnZp5CPcH7F0dhiZkE1P0j0TQ/edit?usp=drive_link&ouid=104610274853675843046&rtpof=true&sd=true

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2.8 Project PPT Preparation: Week 8

- Designed a presentation aligned with selected SDGs, including**:**
- Title Slide – Project title, team members, SDG background
- Problem Statement – Clear, relatable, 2–3 lines
- SDG Alignment – Icons and explanation of contribution
- AI Solution Overview – Flowcharts, input-process-output logic
- Target Audience & Tools/Tech Stack – Python, ChatGPT / Gemini API, Botpress, etc.
- Expected Outcome & Impact – Bullet points, mockups, visuals
- Demo/Prototype Slide – Screenshots or wireframes (if available)

CHAPTER 3: LEARNINGS

During the Lenovo LEAP AI Internship, I gained significant exposure to the foundational and practical aspects of **Artificial Intelligence (AI)** and **Machine Learning (ML)**. The structured weekly modules, hands-on assignments, and project-based tasks offered me a clear understanding of how AI concepts are applied in real scenarios. I learned to plan AI workflows, create simple chatbots, prepare concept notes, and present project ideas through PPTs. This exposure helped me build confidence in approaching AI problems step by step.

3.1 Innovation in Task Handling

While completing the weekly assignments and deliverables, I found opportunities to be innovative in the way I approached tasks. For example, while preparing the chatbot task and PPT assignments, I organized my content more efficiently by using templates and planning the structure early.

This reduced the time required for each submission and helped maintain clarity and quality. I also developed a personal workflow where I watched the weekly videos first, took summary notes, and then applied the concepts practically. This improved my overall productivity and made the tasks easier to handle.

3.2 Personal Learning About Myself

The internship helped me discover several things about my own abilities and working style. I realized that I am capable of learning independently through online modules and adapting to new technical concepts even when they seem challenging at first. Managing weekly deadlines taught me discipline, time-management, and consistency.

I also understood that I enjoy tasks that involve creativity such as designing chatbots, preparing presentations, and developing AI project ideas. At the same time, I recognized areas where I need to improve, such as understanding advanced ML algorithms and being more confident with coding aspects of AI.

Overall, the practical exposure during this internship strengthened my technical knowledge, improved my workflow, and gave me a clearer understanding of my strengths, interests, and growth areas within the field of Artificial Intelligence.

CHAPTER 4: CHALLENGES

During the Lenovo LEAP AI Internship, I encountered certain challenges that helped shape my learning experience and contributed to my overall growth. These challenges mainly arose from being introduced to new technical concepts and the responsibility of completing weekly tasks within the given time frame.

4.1 The Activity That I Was Unaware About

One major challenge was working on activities and concepts that I had no prior experience with. Learning about Machine Learning, AI workflows, and creating chatbots involved ideas and tools that were completely new to me. Understanding these topics required extra effort, such as rewatching weekly modules and exploring additional resources to fully grasp the concepts. Initially, tasks like preparing concept notes, developing chatbots, and planning AI-based project ideas felt difficult because I was unfamiliar with the technical process behind them.

4.2 Meeting the Deadlines

Another challenge I faced was completing the weekly submissions such as PPTs, chatbot tasks, and video-based assignments within the deadlines provided. Balancing these deliverables alongside my academic work required strong time management. Some weeks had more tasks than others, which made staying on schedule difficult. This pushed me to develop better planning habits, divide tasks across the week, and remain consistent with my progress to ensure timely completion.

Appendix : 1 : Sample of work done

All Assignments Drive Link Week 1 - 8
https://drive.google.com/drive/folders/1rg1bCiGGDaVWuW3eOH-MeEBJtAixDoFs?usp=drive_link

Screenshots of the work

Week 2

Link of the Case study
https://drive.google.com/file/d/1VSc8S6Sgkm1410jE2g2sggaO3BvM-umS/view?usp=drive_link

[4]
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
```

[9]
```
data = pd.read_csv('house_pricing.csv')
X = data[['area']]
Y = data['pricing']
data.head()
```

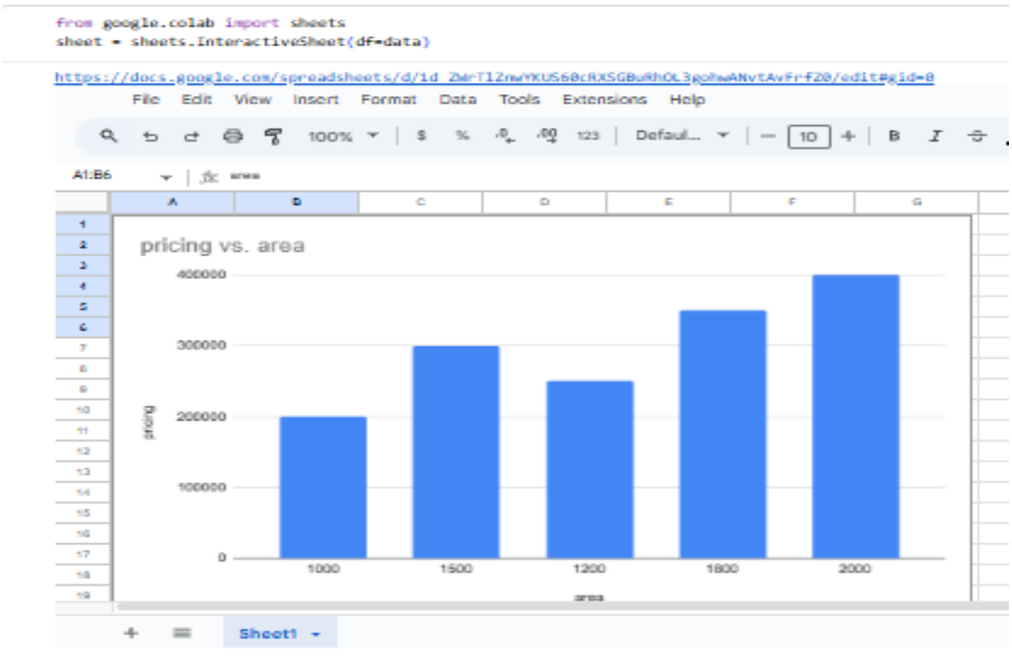
	area	pricing
0	1000	200000
1	1500	300000
2	1200	250000
3	1800	350000
4	2000	400000

```
X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2, random_state=42)

print("Training set shape:", X_train.shape, y_train.shape)
print("Testing set shape:", X_test.shape, y_test.shape)
```

```
Training set shape: (4, 1) (4,)
Testing set shape: (1, 1) (1,)
```

```
data = {'area': [1000, 1500, 1200, 1800, 2000],
        'pricing': [200000, 300000, 250000, 350000, 400000]}
dummy_df = pd.DataFrame(data)
dummy_df.to_csv('house_pricing.csv', index=False)
```



Event Assistant Report of the Chatbot (Week 6)

https://drive.google.com/file/d/1OHTSwQojnngptOWqqw1HDYLBYTbfOf_C/view?usp=drive_link

Concept Notes (Week 7)

https://docs.google.com/presentation/d/13j-EgjFmnZp5CPcH7F0dhiZkE1P0j0TQ/edit?usp=drive_link&ouid=104610274853675843046&rtpof=true&sd=true

Project PPT(Week 8)

https://drive.google.com/file/d/1PWXcsA_sKCWYXbg57TB9TxNdYViROnfo/view?usp=drive_link

Conclusion

The 8-week AI Internship at Lenovo Leap: NextGen Scholar Program was a transformative experience that significantly enhanced my understanding and practical application of Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, and Generative AI. Through structured modules, hands-on projects like building a Linear Regression model, designing a specialized AI assistant, and developing a chatbot using Botpress.

REFERENCES

1. **Lenovo Leap – NextGen Scholar Program (2025)**. - Artificial Intelligence Internship Modules, Training Materials, and Project Guidelines. Lenovo India Pvt. Ltd. https://leapnextgenscholar.in/
2. **Chatgpt** - https://chatgpt.com/
3. **GMFC College WorkShop -** The Workshop helped us to lead to the program.
4. **Google Colab** - https://colab.research.google.com/
5. **Botpress** - https://botpress.com/
6. **Google docs** - https://docs.google.com/document/u/0/
7. **Google Drive Folders -** https://drive.google.com/drive/folders/1rg1bCiGGDaVWuW3eOH-MeEBJtAixDoFs?usp=drive_link
8. **UNDP** – United Nations Development Programme. Sustainable Development Goals (SDGs) Framework. Available at: https://www.undp.org/sustainable-development-goals
9. **Lenovo Official website**- https://www.lenovo.com/